INCH-POUND

MIL-C-14750B (AR)

18 JUNE 1989
SUPERSEDING
MIL-C-14750A (PA)
23 November 1973

MILITARY SPECIFICATION

CARTRIDGE, 60MM, HE, M49A4
LOADING, ASSEMBLING AND PACKING

This Specification is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the loading, assembling and packing of high explosive cartridge designated as the M49A4 for use with the 60mm Mortar.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specifications for
MIL-P-60316 - Primer, Percussion, M32, Loading, Assembling and Packing

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament, Munitions, and Chemical Command, Attn. AMSMC-QA, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1310

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-410 - Nondestructive Testing Personnel Qualification and Certification (Eddy Current, Liquid Penetrant,
MIL-STD-453 - Inspection, Radiographic
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS (See 6.4)

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

PRODUCT AND PACKAGING DRAWINGS

9220014 - Container, Ammunition, Fiber, M576, for 60MM, HE, Cartridge, M49A4, in Jungle Wrap
9220015 - Box, Packing Ammunition, for 60MM Cartridge HE, M49A4, in Jungle Wrap
9220179 - Cartridge, 60MM, HE, M49A4
9234844 - Container, Metal, Ammunition PA 60 for 60MM Mortar Ammunition (Marine Corps Pack)
9234845 - Box, Wirebound, Packing Ammunition for 60MM Mortar Ammunition (Marine Corps Pack)

9254735 - Container, Ammunition, Fiber PA59 for 60MM Cartridge, HE or 60MM Cartridge, TP (Marine Corps Pack)

INSPECTION EQUIPMENT DRAWINGS

9246386 - Inspection Equipment for Measuring Leakage of Metal Ammunition Containers
7256348 - Alignment Gage

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(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific acquisition functions should be obtained from contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. Material shall be in accordance with applicable drawings and specifications.

3.1.1 Clean explosive scrap. Clean scrap (see 6.5) produced in regular operations may be remelted and reused provided that it complies with the requirements for the grade of high explosive specified, except for form. Scrap from floor sweepings, ventilation, and suction apparatus (other than from drilling operations specifically controlled to prevent contamination of explosive) shall not be used.

3.2 Cartridges. The cartridge shall comply with all requirements specified on Drawing (dwg.) 9220179 and with all requirements specified in the applicable specifications.

3.3 Surface cavitation. Cavities formed due to chipped or broken explosive from the sidewalls of the fuze well shall not exceed in aggregate more than 1/4 of the circumference. The maximum (max.) dimension of any visible cavity in the faced (bottom) surface of the fuze well shall not exceed 1/4 inch in any direction and the total area of all cavities shall not exceed 80 percent of the faced surface area. No individual point of explosive shall extend above the flat surface on which the liner would rest. (Cavities having a max. dimension of 1/16 inch or less shall be disregarded in the computation.)

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3.4 Foreign matter in projectile cavity. There shall be no foreign matter in the projectile cavity prior to high explosive pouring. Solid explosive shall not be present.

3.5 High explosive on threads. Threads shall be free of high explosive.

3.6 High explosive on projectile body. Exterior surface of the projectile body shall be free of high explosive.

3.7 Cavitation in high explosive charge.

3.7.1 General. Cavities having a maximum (max.) projected dimension of 3/32 inch or less shall be disregarded provided that they are separated by 1/2 inch or more of solid explosive.

3.7.2 Maximum projected areas and dimensions of cavities. The aggregate of the projected area of all cavities shall not exceed 1 square inch and no individual cavity shall exceed 1/2 square inch. Cavities located in the area of the charge which extends from the base of the fuze well to 2.5 inches below the base of the fuze well shall not exceed 3/4 inch in any direction and their aggregate projected area shall not exceed 5/8 square inch.

3.7.3 Piping cavity. A piping cavity on the longitudinal axis of the projectile shall be permitted provided that its projected length does not exceed 1 1/2 inches and its width does not exceed 1/4 inch. The projected area of such a cavity shall not be included when determining cavity area in the charge.

3.7.4 Cracked charges and annular rings. Not more than four transverse cracks shall be permitted in any charge. The cracks shall be no wider than 1/32 inch. Longitudinal cracks shall be disregarded. The presence of annular rings with a projected width of 1/16 inch max. and extended from the shell wall 1/4 inch max. into the charge shall not be cause for rejection.

3.7.5 Base separation. The base separation from the inside of the rear of the shell to the base of the explosive shall be not more than 1/32 inch.

3.8 Ballistic performance (proving ground). The cartridge shall comply with the following requirements when tested as specified in 4.4.3.2.

3.8.1 Misfire. The cartridge shall not fail to fire.

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3.8.2 Safety. There shall be no premature burst or indication of metal parts separation in the mortar tube or within 100 meters travel from the gun tube when fired at any charge zone.

3.8.3 Impact functioning. The cartridge shall function high order upon impact.

3.8.4 Horizontal range.

3.8.4.1 Short rounds. No cartridge shall have a horizontal range less than 80 percent of the average horizontal range.

3.8.4.2 Range precision. The standard deviation in horizontal range shall be no greater than the maximum (max.) lot values shown in Table I for the increment charge indicated.

TABLE I Range Precision

Increment Charge Zone	Max. Lot Standard Deviation (Meters)
0	17
4	44

3.9 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.3).

3.10 Workmanship. All parts and assemblies shall be fabricated and loaded in a thorough workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, rust and other foreign matter. The method used shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents. All required markings shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of

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the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.1.1 Responsibility for compliance. All items must meet all requirements of Section 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Alternate inspection provisions. Alternative inspection procedures, methods or equipment (including tool inspection, statistical process control, substitute sampling plans, etc.) may be used by the contractor when they provide, as a minimum the level of quality assurance required by the provisions specified herein. Prior to applying such alternative inspection procedures, methods or equipment, the contractor shall describe them in a written proposal submitted to the Procuring Contracting Officer for evaluation and approval by the Government. When required, the contractor shall demonstrate that the effectiveness of the proposed alternative(s) is equal to or better than the quality assurance provisions herein. In case of dispute as to whether certain procedures of the contractor's proposed alternative(s) provide equal assurance, the provisions of this specification shall apply.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection

b. Quality Conformance Inspection

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4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the components in the quantities described in Table II.

4.3.2 Inspections to be performed. See MIL-A-48078 and Table II specified herein.

4.3.3 Rejection. See MIL-A-48078 and the first article ballistic test shall be rejected per Table III criteria.

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TABLE II. First article inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-C-14750B (AR)

PARAGRAPH	TITLE: Cartridge, Assembly and Container		SHEET 1 OF 1		DRAWING NUMBER: See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Cartridge (Dwg. 9220179)				
	Examination of Defects	25		3.2	4.4.2.4/4.4.2.5
	Ballistic Test (Proving Ground)	80		3.8	4.4.3.2
	Metal Ammunition Container (Dwg. 9234844)				
	Air Leakage Test	5		3.2	4.5.3

NOTES:

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with lot formation provisions of MIL-A-48078. In addition, each inspection lot of cartridges shall contain:

a. Metal parts and metal parts assemblies of one lot interfix number from one manufacturer.

b. Loaded fuzes of one lot interfix number from one manufacturer.

c. Loaded ignition cartridges of one lot interfix number from one manufacturer.

d. Fin assemblies of one lot interfix number from one manufacturer.

e. Loaded primers of one lot interfix number from one manufacturer.

f. Propellant increments of one lot interfix number from one manufacturer.

g. Composition B of one lot interfix number from one manufacturer.

h. Fin extensions of one lot interfix number from one manufacturer.

4.4.2 Examination. See MIL-A-48078. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Body Assembly, Prior to Loading	1 OF 1	9207926
			NEXT HIGHER ASSEMBLY 9220179

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Foreign matter in charge cavity		100%	3.4	Visual
Major					
101	Interior protective coating with bare spot or blister		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-14750B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Body Loading Assembly, Prior to Assembly of Closing Plug or Fuze				9207926
					NEXT HIGHER ASSEMBLY 9220179
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	High explosive on projectile body		100%	3.6	Visual
Major					
101	Cavitation		(a)	3.7	Visual/ 4.5.1
102	Depth of fuze well		0.40%	3.2	Gage
103	Diameter of fuze well		0.40%	3.2	Gage
104	Distance to explosive from forward face		0.40%	3.2	Gage
105	Foreign matter in fuze well		0.40%	3.2	Visual
Minor					
201	Excessive surface cavitation		0.65%	3.3	Visual
202	High explosive on threads		0.65%	3.5	Visual
203	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES: (a) The sampling plan and procedure shall be in accordance with MIL-STD-1235 using CSP-2 plan having the following parameters; i value shall be 144, f value shall be 1/24 and s value shall be 1800.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-14750B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Body Loading Assembly, After Assembly of Closing Plug	1 OF 1	9207926
			NEXT HIGHER ASSEMBLY 9220179

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Total length, max.		0.40%	3.2	Gage
Minor					
201	Protective coating with total damaged area in excess of 1/4 square inch		0.65%	3.2	Visual
202	Marking misleading, missing or unidentifiable including incorrect color code		0.65%	3.2	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-14750B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.4	Cartridge				9220179
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Primer head above flush with rear surface of fin cartridge container		100%	3.2	Gage
2	Pull wire missing or not fully seated		100%	3.2	Gage
3	Fin damaged or distorted		100%	3.2	Gage
4	Ignition cartridge missing or inverted		100%	3.2	Visual
5	Propellant increment missing		100%	3.2	Visual
Major					
101	Profile and alignment		100%	3.2	Gage (see 7256348)
102	Propellant increment cut or damaged		100%	3.2	Visual
103	Fin fails to meet assembly torque		0.40%	3.2	Gage
104	Fuze fails to meet assembly torque		0.40%	3.2	Gage
105	Total length, max.		0.40%	3.2	Gage
106	Any component or operation missing		0.40%	3.2	Visual
107	Primer stake missing or thread sealing compound does not cover a minimum of one half the number of male primer threads for a full 360 degrees (after disassembly as applicable)		0.40%	3.2	Visual
108	Sealing compound bead improperly located, missing or not full 360 degrees in base of cartridge housing cavity (prior to insertion of ignition cartridge)		0.40%	3.2	Visual

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.4	Cartridge				9220179
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
109	Sealing compound bead improperly located, missing or not full 360 degrees on red end of ignition cartridge (prior to seating of primer into fin cavity)		0.40%	3.2	Visual
Minor					
201	Protective coating with total damaged area in excess of 1/4 square inch		0.65%	3.2	Visual
202	Marking misleading, missing or unidentifable, including incorrect color code		0.65%	3.2	Visual
203	Fuze not fully seated		0.65%	3.2	Visual
	Proving Ground Test				
	Production ballistic acceptance	80/48		3.8	4.4.3.2/4.5.2

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.5	Unsealed Fiber Container (Select Appropriate dwg.)				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Ring support missing		100%	5.1	Visual
Minor					
201	Glue or asphalt on round		0.65%	5.1	Visual
202	Cartridge improperly assembled into container		0.65%	5.1	Visual
203	Round cannot be removed by hand		0.65%	5.1	Visual/Manual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-14750B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.6	Sealed Fiber container (For Army Use)				9220014
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Container damaged or cut through all asphalt impregnated layers		0.40%	5.1.1	Visual
102	Tape incomplete or badly wrinkled or missing		0.40%	5.1.1	Visual
103	Metal end loose or distorted		0.40%	5.1.1	Visual
104	Cushion or filler missing		0.40%	5.1.1	Visual
Minor					
201	Sealing strip improperly applied		0.65%	5.1.1	Visual
202	Marking missing, misleading or unidentifiable		0.65%	5.1.1	Visual
203	Contents loose		0.65%	5.1.1	Manual
204	Gap between cover and body excessive (exceeds 1/8 inch)		0.65%	5.1.1	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-14750B (AR)

PARAGRAPH	TITLE		SHEET	DRAWING NUMBER
4.4.2.6.1	Sealed Fiber Container (Marine Corps Use)		1 OF 1	9254735
				NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Container damaged or cut through all asphalt impregnated layers		0.40%	5.1.1	Visual
102	Tape incomplete or badly wrinkled or missing		0.40%	5.1.1	Visual
103	Metal end loose or distorted		0.40%	5.1.1	Visual
Minor					
201	Sealing strip improperly applied		0.65%	5.1.1	Visual
202	Marking missing, misleading or unidentifiable		0.65%	5.1.1	Visual
203	Contents loose		0.65%	5.1.1	Manual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-14750B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.7	Fiber Container Wrapped in Barrier Material Prior to Waxing-Jungle Wrap (For Army Use)	1 OF 1	9220014
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Barrier material improperly assembled		0.40%	5.1.1	Visual
102	Barrier material ends improperly folded or taped		0.40%	5.1.1	Visual
Minor					
201	Marking missing or incomplete		0.65%	5.1.1	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA-(D) Form 160, 1 Aug 83, which may not be used.

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.7.1	Fiber Container After Waxing (Jungle Wrap) (Army Use)				9220014
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Wax coating incomplete or damaged		0.40%	5.1.1	Visual
Minor					
201	Marking missing or unidentifiable		0.65%	5.1.1	Visual
202	Kraft paper improperly placed on container		0.65%	5.1.1	Visual
203	Tear tab not visible		0.65%	5.1.1	Visual

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.8	Metal Ammo Box (Prior to Closing) (Marine Corps Use)	1 OF 1	9234844
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Either support missing or incorrect		0.40%	5.1.1	Visual
102	Incorrect number of assemblies		0.40%	5.1.1	Visual
Minor	None defined				

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.8.1	Metal Ammo Box-Closed (Marine Corps Use)				9234844
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Air leakage test		100%	5.1.1	4.5.3
102	Hardware improperly engaged		0.40%	5.1.1	Visual
103	Box dented or deformed		0.40%	5.1.1	Visual
104	Box punctured		0.40%	5.1.1	Visual
Minor					
201	Marking missing, misleading or unidentifiable		0.65%	5.1.1	Visual
202	Contents loose		0.65%	5.1.1	Manual
203	Bare spot on exterior		0.65%	5 1.1	Visual

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.9	Sealed Wood Packing Box (Select Appropriate dwg.)				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Contents exposed		0.40%	5.2.1	Visual
102	DOD symbol missing or incorrect		0.40%	5.2.1	Visual
103	Board broken or split		0.40%	5.2.1	Visual
104	Hardware or strapping missing broken or loose		0.40%	5.2.1	Visual/Manual
Minor					
201	Marking missing, misleading or unidentifiable		0.65%	5.2.1	Visual
202	Hardware improperly engaged		0.65%	5.2.1	Visual
203	Car seal missing, unsealed or improperly positioned		0.65%	5.2.1	Visual
204	Contents loose		0.65%	5.2.1	Visual/Manual
205	Handle missing or insecure		0.65%	5.2.1	Visual/Manual

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Tables.

4.3.1 Check test for possible deterioration of primer, precussion M32 (see MIL-P-60316). If the total elapsed time between original acceptance of the M32 primer lot and the assembly of that lot into complete round cartridges exceeds two years, or if the M32 primers have been subjected to adverse conditions, however brief, at any time since previous test, the applicable lot shall be subjected to and must satisfactorily pass the check test for deterioration specified in MIL-P-60316 before the M32 primer lot is assembled into the cartridges.

4.4.3.2 Proving ground test. The cartridge shall be observed for compliance with the requirements as specified in Table III.

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TABLE III Round Performance

Defect	Defect Classification	Accept	Reject
(a) Misfires (see 3.8.1)	Major	1	2
(b) Safety (see 3.8.2)	Critical	0	1
(c) Impact functioning (see 3.8.3)	Major	3	4
(d) Short rounds (see 3.8.4.1) less than 80 percent of average horizontal range but not less than 60 percent of average horizontal range.	Special	0	1
(e) Short rounds less than 60 percent of average horizontal range.	Critical	0	1
(f) Range standard deviation (see 3.8.4.2)	Major	-	-

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4.4.3.2.1 Initial production test. Beginning with the first lot produced and continuing until three consecutive lots have complied with the applicable requirements, a sample of 80 cartridges shall be slected from each lot for this test. See Table III for acceptance criteria. In addition, if the standard deviation in horizontal range for the sample cartridges fired at charge zero or charge four exceeds the applicable requirement of Table I, see (3.8.4.2) the lot shall be rejected. The test shall be performed as specified in 4.3.2.

4.4.3.2.2 Production test. After three consecutive lots have met the criteria of 4.2.3.3.1 a sample of 48 cartridges shall be selected from each lot for this test. See Table II for acceptance criteria. In addition, if the standard deviation in horizontal range for the sample cartridges fired at charge, zone zero or charge zone four exceeds the applicable max. limit specified in Table I (see 3.8.4.2) the lot shall be rejected. The test shall be performed as specified in 4.5.2.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed in this specification is identified either directly or by reference, in the "Paragraph Reference/Inspection Method" column of the Quality Conformance Inspection tables herein. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.2 herein.

4.5 Methods of inspection.

4.5.1 Determination of cavities. The sample cartridges (HE filled) shall be subjected to radiographic examination. The radiographic beam shall form a 90° angle with the longitudinal axis of the projectile. Examinations of radiograph shall be made to determine compliance with applicable requirements. Radiographic equipment, operations and procedures shall be

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qualified in accordance with MIL-STD-453. In addition, the sensitivity of the radiographic technique shall be 2.0 percent, and the photographic density of the film used shall be no less than 1.25, as determined in accordance with 4.5.1.2. In the event an original radiograph leave doubt as to acceptability, the cartridge shall be re-radiographed at an angle 90° from that of the original exposure and in the same plane.

4.5.1.1 Sensitivity check of radiographic technique. The sensitivity of radiographic technique shall be determined by the means of a penetrameter fabricated as specified herein. Location of the penetrameter hole shall be 3 3/4 inches measured from base shoulder of projectile body towards fuze nose threads. The projectile, with attached penetrameter, shall be positioned between the radiographic source and the radiographic film with the penetrameter facing the radiographic source. When radiographing a group of projectiles on a rotating ring, the penetrameter, taped to a projectile, may be used to represent the group when checking sensitivity. The sensitivity of radiographic technique shall be considered satisfactory when the image of the penetrameter, including the holes, is sharply defined on the radiograph. The penetrameters shall be made of steel having an approximate specific gravity of 7.8. Penetrameter thickness shall be .020 plus or minus (+) .010 inches. Each penetrameter shall have a circular hole with a diameter twice that of the penetrameter thickness. For further details on construction of the penetrameters, see MIL-STD-453. Frequency of sensitivity check shall be no less than one in each fifty (50) radiographs.

4.5.1.2 Check of photographic density of film. The photographic density of the film shall be determined on the film in the region of the maximum equivalent thickness to be radiographed. Film density shall be checked each hour of operation, or fraction thereof, for each film processor. Greater frequency shall be required for any period during which dificulty is encountered in establishing the required densities. To judge densities, it is recommended that standards be used. Comparison of the film with standards may be made visually; however, the use of densitometer is recommended.

4.5.1.3 Personnel qualification and certification. Personnel performing the radiographic examination in accordance with paragraph 4.5.3 shall meet the qualification and certification requirements defined in MIL-STD-410 as follows:

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4.5.1.3.1 Set up and calibration. Level II.

4.5.1.3.2 Operator. Level I.

4.5.1.3.3 Documentation. The contractor will prepare and submit for approval, in accordance with data item description DI-S-3628 and the DD Form 1423, his detailed procedures for qualification and certification of personnel performing the non-destructive testing procedures required herein.

4.5.2 Proving ground test. The test shall be performed at a Government proving ground, in accordance with the applicable Acceptance Test Procedure. The cartridge shall be fired from a 60mm mortar for which the round is standard. One half of the sample cartridges shall be fired with an ignition cartridge (charge zone zero) and one half of the sample cartridges shall be fired with an ignition cartridge and four propellant increments (charge zone four). The individual velocities, impact ranges and fuze action shall be obtained and recorded. The standard deviation and the arithmetic average of the impact ranges shall be calculated and recorded.

4.5.3 Leakage test (for Marine Corps pack only). Each box shall be tested using a partial vacuum system as per envelope dwg. 9246386 utilizing a chamber with minimal clearance between the exterior of the box and interior of the chamber.

5. PACKAGING

5.1 Preservation and packaging. This specification contains two alternate packaging instructions one for Army and the other for Marine Corps. Section 2, 4, and 5 are affected by the alternate packaging. Selection must be made of appropriate requirements for packaging and associated inspections as they apply to Army or Marine Corps.

5.1.1 Level A. The cartridges shall be preserved and packaged in accordance with dwg. 9220014 for Army or dwg. 9254735 and 9234844 for Marine Corps.

5.1.2 Level B. Same as level A.

5.1.3 Level C. Same as level A.

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5.2 Packing.

5.2.1 Level A. The cartridges in fiber containers shall be packed in accordance with dwg. 9220015 for Army or dwg. 9234845 for Marine Corps.

5.2.2 Level B. Same as level A.

5.2.3 Level C. Same as level A.

5.3 Marking. Select appropriate paragraph, 5.3.1 for Army rounds or 5.3.2 for Marine Corps rounds.

5.3.1 Marking-Army. Marking of the wood box shall be in accordance with dwg. 9220015. The fiber container shall be marked in accordance with dwg. 9220014.

5.3.2 Marking-Marine Corps. Marking of the fiber container shall be in accordance with dwg. 9254735. The metal container shall be marked in accordance with dwg. 9234844. The wood box shall be marked in accordance with dwg. 9234845.

5.3.3 Level B. Same as level A.

5.3.4 Level C. Same as level A.

6. NOTES

6.1 Ordering data. See MIL-A-48078 (AR).

6.2 Submission of inspection equipment for design approvals. See MIL-A-48078. Submit designs as required to Commander, U.S. Army ARDEC, ATTN: AMSMC-QAT-I (D), Bldg 62, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply.

6.3 Submission of test data. One copy of proving ground ballistic acceptance test data (i.e., firing record) shall be forwarded to the following:

Commander
U.S. Army Research, Development and Engineering Center
ATTN: AMSMC-QAT-B (D) & SMCAR-FSA-IM
Picatinny Arsenal, NJ 07806-5000

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6.4. Drawings. Drawings listed in Section 2 of this specification under the headings U.S. Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal and U.S. Armament Research and Development Command (ARRADCOM) drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.5 Clean explosive scrap. Clean scrap is defined as scrap formed in kettles, pails, risers and loading machines used in the pouring and cooling operations, and scrap formed during the drilling operation.

6.6 Measurement of ballistic samples. The following measurements shall be taken after all mechanical operations have been performed on the ballistic samples.

6.6.1 Bourrelet diameter. Two measurements shall be taken at both the front and rear of the bourrelet, 90 degrees apart.

6.6.2 Concentricity of fin extension. The projectile shall be clamped in rollers on the bourrelet diameter and the fin extension concentricity (total indicator reading) measured at approximately 1/4 inch forward of the fin.

6.6.3 Straightness of fin blades. The projectile shall be clamped in rollers on the bourrelet diameter and the straightness (total indicator reading) measured along the length of each blade.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

6.7 Subject term (key word) listing.

Body assembly
Body loading assembly
Cartridge
Cavitation
High explosive
Percussion primer

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6.8 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph No.	Data Requirement Title	Applicable DID No.	Option
4.5.1.3.3	Nondestructive Inspection Plan	DI-S-3628	-

(Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1310-A439)

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